

Timothy C. Peter
Site Vice President- JAF

JAFP-23-0045
August 14, 2023

United States Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, D.C. 20555-0001

James A. FitzPatrick Nuclear Power Plant
Renewed Facility Operating License No. DPR-059
NRC Docket No. 50-333

Subject: LER: 2023-002-01, Reactor Protection System Electric Power Monitoring
System Trip Caused Primary Containment Isolation

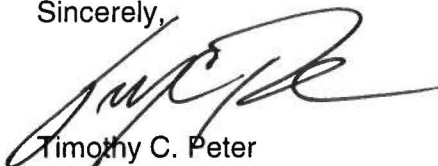
Dear Sir or Madam:

This report is being submitted pursuant to 10 CFR 50.73(a)(2)(iv)(A).

There are no new regulatory commitments contained in this report.

Questions concerning this report may be addressed to Mr. Mark Hawes, Regulatory Assurance,
at (315) 349-6659.

Sincerely,



Timothy C. Peter
Site Vice President

TCP/MH

Enclosure: LER: 2023-002-01, Reactor Protection System Electric Power Monitoring
System Trip Caused Primary Containment Isolation

cc: USNRC, Region I Administrator
USNRC, Project Manager
USNRC, Resident Inspector
INPO Records Center (IRIS)

NRC FORM 366 (03-14-2023)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 08/31/2023				
LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/) Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.											
1. Facility Name James A. FitzPatrick Nuclear Power Plant					☒ 050 ☐ 052		2. Docket Number 05000333		3. Page 1 OF 4		
4. Title Reactor Protection System Electric Power Monitoring System Trip Caused Primary Containment Isolation											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050 Docket Number N/A N/A	
04	14	2023	2023	002	01	08	14	2023	N/A	☐ 052 Docket Number N/A N/A	
9. Operating Mode 1					10. Power Level 100						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify in Abstract below or in NRC Form 366A).											
12. Licensee Contact for this LER											
Licensee Contact Mr. Mark Hawes, Regulatory Assurance									Telephone Number (Include Area Code) 315-349-6659		
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS		
D	JE	EC	O026	N							
14. Supplemental Report Expected						15. Expected Submission Date			Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission date)									
Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)											
On April 14, 2023, at 05:11, James A. FitzPatrick Nuclear Power Plant (JAF) received indication of a half-scam on the "B" side Reactor Protection System (RPS). The cause was determined to be an overvoltage condition that tripped the Electrical Protection Assembly (EPA) for the "B" RPS Motor Generator (MG) Set. The same condition occurred on May 25, 2023, at 23.59. These events caused invalid primary containment isolation signals in multiple systems, reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A). The cause was corrosion buildup in the voltage adjust potentiometer which led to the overvoltage trip of the EPA. Corrective actions included replacing the voltage adjust potentiometer.											

NRC FORM 366A (03-14-2023)	U.S. NUCLEAR REGULATORY COMMISSION LICENSEE EVENT REPORT (LER) CONTINUATION SHEET	APPROVED BY OMB: NO. 3150-0104 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.	EXPIRES: 08/31/2023
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(See NUREG-1022, R.3 for instruction and guidance for completing this form <http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

1. FACILITY NAME James A. FitzPatrick Nuclear Power Plant	☒ 050 ☐ 052	2. DOCKET NUMBER 05000333	3. LER NUMBER		
			YEAR 2023	SEQUENTIAL NUMBER - 002	REV NO. - 01

NARRATIVE

Background

Reactor Protection System (RPS) [EIS: JE] Electric Power Monitoring System is provided to isolate the RPS bus from the motor generator (MG) set or an alternate power supply in the event of overvoltage, undervoltage, or underfrequency. This system protects the loads connected to the RPS bus against unacceptable voltage and frequency conditions and forms an important part of the primary success path of the essential safety circuits. Some of the essential equipment powered from the RPS buses includes the RPS logic, scram pilot valve solenoids, and various valve isolation logic.

RPS electric power monitoring assembly will detect any abnormal high or low voltage or low frequency condition in the outputs of the two MG sets or the alternate power supply and will de-energize its respective RPS bus, thereby causing all safety functions normally powered by this bus to de-energize. (Safety functions powered by the RPS buses deenergize to actuate.)

Two redundant Class IE circuit breakers are connected in series between each RPS bus and its MG set, and between each RPS bus and its alternate power supply. Each of these circuit breakers has an associated independent set of Class IE overvoltage, undervoltage, and under frequency sensing logic. Together, a circuit breaker and its sensing logic constitute an electric power monitoring assembly. If the output of the inservice MG set or alternate power supply exceeds predetermined limits of overvoltage, undervoltage, or underfrequency, a trip coil driven by this logic circuitry opens the circuit breaker, which removes the associated power supply from service.

Event Description

On April 14, 2023, at 05:11 when James A. FitzPatrick Nuclear Power Plant (JAF) was at 100% power, the station received indication of a half-scam on "B" side RPS. AOP-60, Loss of RPS Bus B Power, was entered and the cause was determined to be a trip of the "B" RPS MG set due to an overvoltage condition. The direct cause of the output breaker tripping was due to an Electric Power Monitoring System overvoltage trip on 71EPA-RPS1B1G, Electrical Protection Assembly (EPA).

On May 25, 2023, at 23:59, when JAF was at 100% power, the station received indication of a half-scam on the "B" side RPS. AOP-60 was entered and the cause was determined to be a trip of the "B" RPS MG set due to an overvoltage condition. The direct cause of the output breaker tripping was due to an Electric Power Monitoring System overvoltage trip on 71EPA-RPS1B1G, EPA.

In both events, the RPS bus power was restored when it was placed on the alternate power supply.

Both events resulted in invalid system actuations of "B" side half scram and Primary Containment Isolation System (PCIS) [JM] isolations. The following systems isolated as a result of the loss of "B" RPS bus: Reactor Water Cleanup [CE], Reactor Building ventilation [NG], "B" Containment Atmosphere Dilution [BB], Torus Vent and Purge, Drywell Equipment and Floor Drain Sumps [WK], "B" Drywell Containment Atmospheric Monitors [IP], Recirculation System Sample Line, and Main Steam Line Drains [SB]. "B" Standby Gas Treatment System [BH] started as designed. These system actuations were not initiated by signals in response to actual plant conditions or parameters satisfying the requirements for initiation of the system.

This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

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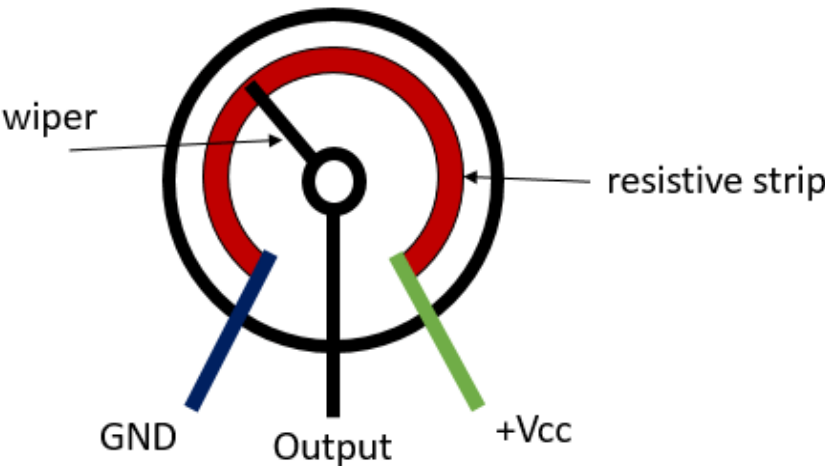
Event Analysis

Following the April 14 “B” RPS MG set trip, initial troubleshooting determined the most likely cause of the event was a degraded voltage regulator. The voltage regulator was replaced and sent to PowerLabs for analysis. The voltage adjust potentiometer “1P” was not replaced. On April 16, 2023, at 22:27, the “B” RPS MG set was restored to service as the RPS bus power supply.

PowerLabs testing concluded that there was no issue with the voltage regulator component in a report dated May 15, 2023; however, they identified the most probable cause to be the voltage adjust potentiometer “1P” component.

After the May 25 trip, a Simpson meter (analog multimeter) was used during troubleshooting and the meter was observed to waiver erratically across the potentiometer full range of movement. The expected response is smooth movement as the potentiometer is moved through its full range. The potentiometer was replaced and sent to PowerLabs for analysis.

The erratic performance of the potentiometer “1P” was determined to be poor electrical connection, likely between the wiper and winding. The most likely cause of the poor electrical connection is oxidation on the wiper and winding mating surfaces. This corrosion introduced an elevated electrical resistance between the wiper and coil. Elevated resistance caused the voltage regulator output to rise, which caused the RPS MG set voltage to rise above the overvoltage trip voltage of the EPA.



Potentiometer wiping preventive maintenance (PM) should detect and prevent failure. The current PM has technicians take an ‘as-found’ resistance reading using an ohmmeter, then wipe the potentiometer, and take an as-left reading. Best practices were determined to be a Simpson meter or similar analog type should be used so the potentiometer resistance can be measured across the entire range to determine if high-resistance spots exist on the potentiometer. Wiping should continue until the high-resistance sections are eliminated and if that is not possible, the potentiometer should be replaced. Based on failure modes team input and an industry benchmark preventative maintenance guidance is being updated to reduce the likelihood of future failures.

Cause

The "B" side half scram, system actuations, and Primary Containment Isolation System (PCIS) isolations were caused by corrosion buildup in the voltage adjust potentiometer which led to an overvoltage trip of the Electrical Protection Assembly (EPA).

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Similar Events

Condition Report – CR-JAF-2012-01491 dated March 14, 2012, “B” RPS bus failed to transfer from alternate power supply to MG Set. Most probable cause is momentary high resistances across the voltage adjust potentiometer “1P”. Corrective actions included revising the PM for the RPS MG sets to include periodic exercise of the potentiometer. This new maintenance practice was ineffective at preventing the event reported in this report.

Corrective Actions

Completed Actions

Replaced voltage adjustment potentiometer “1P”.

Planned Actions

Revise “A” and “B” RPS MG set voltage adjustment potentiometer PM to incorporate the use of analog multimeter and include expected response.

FAILED COMPONENT IDENTIFICATION:

Manufacturer:	Ohmite
Manufacturer Model Number:	RHS500E
Manufacturer Code:	O026
Component Code:	EC
FitzPatrick Component ID:	Voltage adjustment potentiometer “1P” is a subcomponent to 71RP-1B(GEN)

Safety Significance

There were no actual nuclear consequences. RPS Electric Power Monitoring System responded as designed to an overvoltage condition to protect the electrical power systems. Then system actuations responded to the loss of power condition as expected without any issue. There was no loss of any safety functions during these events.

References

Issue Report - IR 04670001, AOP-60 entry due to B side half scram, dated April 14, 2023

Issue Report - IR 04680732, Entry into AOP-60, dated May 26, 2023